

# MMTableBench: A Multi-level Multimodal Benchmark for Reasoning and Layout Complexity in TableQA

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## Abstract

Tables serve as a core format for representing structured data on the web, as their two-dimensional layouts effectively encode complex inter-entity relationships. However, real-world web tables often feature heterogeneous structures and rich semantics. Accurately interpreting such tables requires not only spatial layout perception but also multi-step reasoning across rows and columns, posing substantial challenges to web intelligence systems. Multimodal large language models (MLLMs) show promise in table question answering (TableQA) by leveraging visual layouts. However, their performance on complex web tables remains uneven, as existing benchmarks often blur the impact of individual difficulty factors, hindering precise capability analysis. To advance TableQA beyond superficial task difficulty and toward interpretable capability modeling, we introduce MMTableBench, a multi-level benchmark that systematically evaluates MLLMs along two fine-grained dimensions: layout complexity and reasoning complexity. By organizing table-question pairs along these axes, MMTableBench facilitates a detailed evaluation of model performance under varying structural and reasoning challenges, while revealing the respective strengths and limitations of multimodal inputs. Our comprehensive analysis shows that state-of-the-art MLLMs continue to exhibit notable limitations when confronted with complex layouts and deep reasoning tasks, underscoring persistent gaps despite the structural advantages offered by visual inputs. MMTableBench thus provides not only a rigorous evaluation framework but also a diagnostic tool for analyzing and interpreting model behaviors, enabling more transparent and explainable progress in multimodal TableQA development. Code and Dataset are available at: <https://github.com/TableBench/MMTableBench>.

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## CCS Concepts

• **Computing methodologies** → **Spatial and physical reasoning**; *Language resources*; *Natural language generation*.

## Keywords

Multimodal Large Language Models, Visual Reasoning, Table Understanding, Table Question Answering

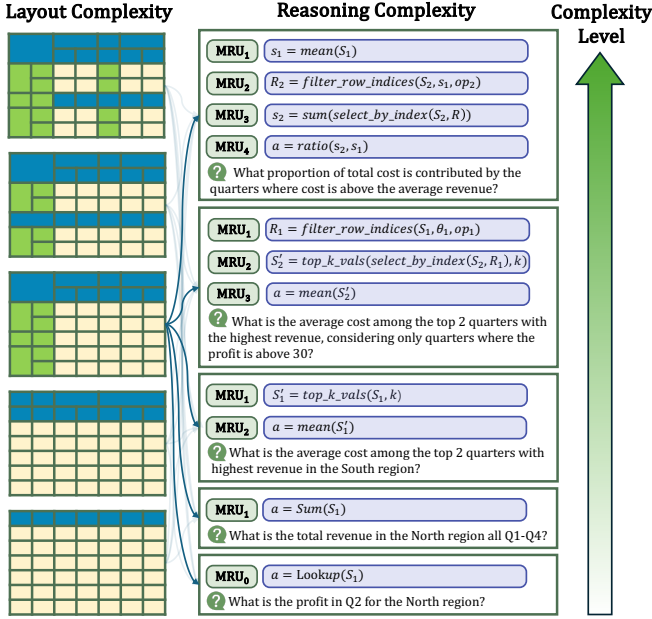
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## 1 Introduction

Tables serve as a foundational medium for organizing structured data on the web, widely used in domains such as e-commerce, finance, and scientific publishing [12]. Their two-dimensional layouts efficiently capture complex inter-entity relationships, making them indispensable to web intelligence systems that aim to understand and reason over online content [9, 16]. However, real-world web tables often appear in rendered formats—such as screenshots, scanned reports, or PDF documents—where the original schema is visually encoded but no longer machine-readable. These visual tables feature diverse structures, including merged cells, hierarchical headers, and spatial alignments, which pose substantial challenges for layout perception and logical inference.

To address these challenges, multimodal large language models (MLLMs) [6, 8, 18] aim to integrate visual perception and symbolic reasoning by jointly processing visual and textual modalities. Although recent advances demonstrate encouraging progress in multimodal TableQA, a critical trade-off remains unresolved: models with strong visual parsing capabilities (e.g., LLaVA variants) [18] often struggle with complex multi-step reasoning, whereas those



**Figure 1: Illustration of the two core dimensions in MMTTableBench.** The left side showcases five representative table structures, from simple to complex layouts. The right side illustrates question-answering examples categorized into five levels of reasoning complexity, where MRUs (Micro Reasoning Units) denote fine-grained reasoning steps. These two dimensions jointly define a comprehensive evaluation space for detailed analysis of model performance.

with advanced reasoning abilities (e.g., GPT-4) [1] frequently misinterpret intricate layouts due to insufficient structural perception.

A key bottleneck in advancing multimodal TableQA is the absence of fine-grained benchmarks that clearly separate layout understanding from reasoning ability. Although recent datasets increasingly incorporate tasks requiring both structural perception and logical inference, they often entangle these aspects. For example, benchmarks like HiTab [7] and RealHiTab [29] commonly pair multi-step reasoning questions with irregular or nested table structures. This conflation obscures the root cause of model errors, making it difficult to determine whether failures arise from reasoning limitations, layout misinterpretation, or both.

**To advance TableQA beyond superficial task difficulty and toward interpretable capability modeling,** we propose a multi-level benchmark, MMTTableBench, which explicitly decouples layout complexity and reasoning depth along two orthogonal axes of evaluation, as illustrated in Figure 1. By separating structural and cognitive challenges, MMTTableBench enables a more interpretable and fine-grained assessment of model capabilities under multimodal conditions. MMTTableBench comprises carefully curated question-table pairs, where each table is extracted from an arXiv scientific paper and provided with both its rendered image and corresponding LaTeX source. To enable interpretable and controllable quantification of both layout and reasoning complexity, we introduce tailored annotation strategies for each dimension. Layout

complexity is categorized along three structural aspects—header hierarchy, cell merging, and composite structure—based on which we establish a principled five-level scoring scheme. For reasoning, all questions are constructed from symbolic reasoning paths composed of Micro Reasoning Units (MRUs), each representing a fine-grained atomic operation such as table lookup or computation. We implement the proposed complexity quantification within a novel annotation framework that aligns visual tables, textual tables, and step-wise reasoning chains with controllable complexity, which facilitates systematic data construction and enables fine-grained control over both structural and reasoning difficulty.

We conduct a comprehensive evaluation of 15 advanced multimodal large language models (MLLMs), covering challenges along both layout complexity and reasoning depth dimensions. This evaluation allows us to systematically diagnose the limitations of existing models and uncover performance bottlenecks under varying structural and cognitive demands. Leveraging MMTTableBench, we further explore the interplay between visual and textual modalities, revealing the extent to which models exhibit cross-modal complementarity. These insights demonstrate the utility of MMTTableBench as a diagnostic and guiding framework for advancing the design of more capable and interpretable multimodal reasoning models.

The contributions are summarized as follows:

- We introduce MMTTableBench, a benchmark that uniquely disentangles layout complexity and reasoning depth as independent evaluation dimensions, enabling interpretable assessment of multimodal table reasoning.
- We introduce a novel annotation framework that enables three-way alignment among visual tables, textual tables, and reasoning chains with controllable complexity, thereby improving the systematic organization and controllability of data construction.
- We conduct extensive experiments on 15 MLLMs and uncover modality-specific strengths and limitations, offering empirical insights into the current bottlenecks of visual-linguistic table understanding.

## 2 Task formulation

The TableQA task aims to generate an answer set  $A$  given a natural language question  $Q$  and a table  $T$ , formally defined as:

$$f : (T, Q) \rightarrow A, \quad (1)$$

where  $A = \{a_1, a_2, \dots, a_k\}$  is a set of values or entities, and  $k \in \mathbb{N}^+$ .

Unlike conventional TableQA settings that represent tables solely in textual form, real-world tables often exhibit complex layouts—such as multi-level headers, merged cells, and irregular structures—that challenge linear serialization. Textual form fails to preserve critical spatial and structural information, which limits the ability of the model to perform accurate reasoning over tabular content.

To address this issue, we redefine the table  $T$  as a multimodal representation integrating both visual and textual modalities:

$$T = \{T_{\text{image}}, T_{\text{text}}\}, \quad (2)$$

where  $T_{\text{image}}$  retains the visual layout, including cell boundaries and spatial organization, while  $T_{\text{text}}$  provides a serialized textual description, expressed in LaTeX. This multimodal formulation enables the model to leverage complementary information from both

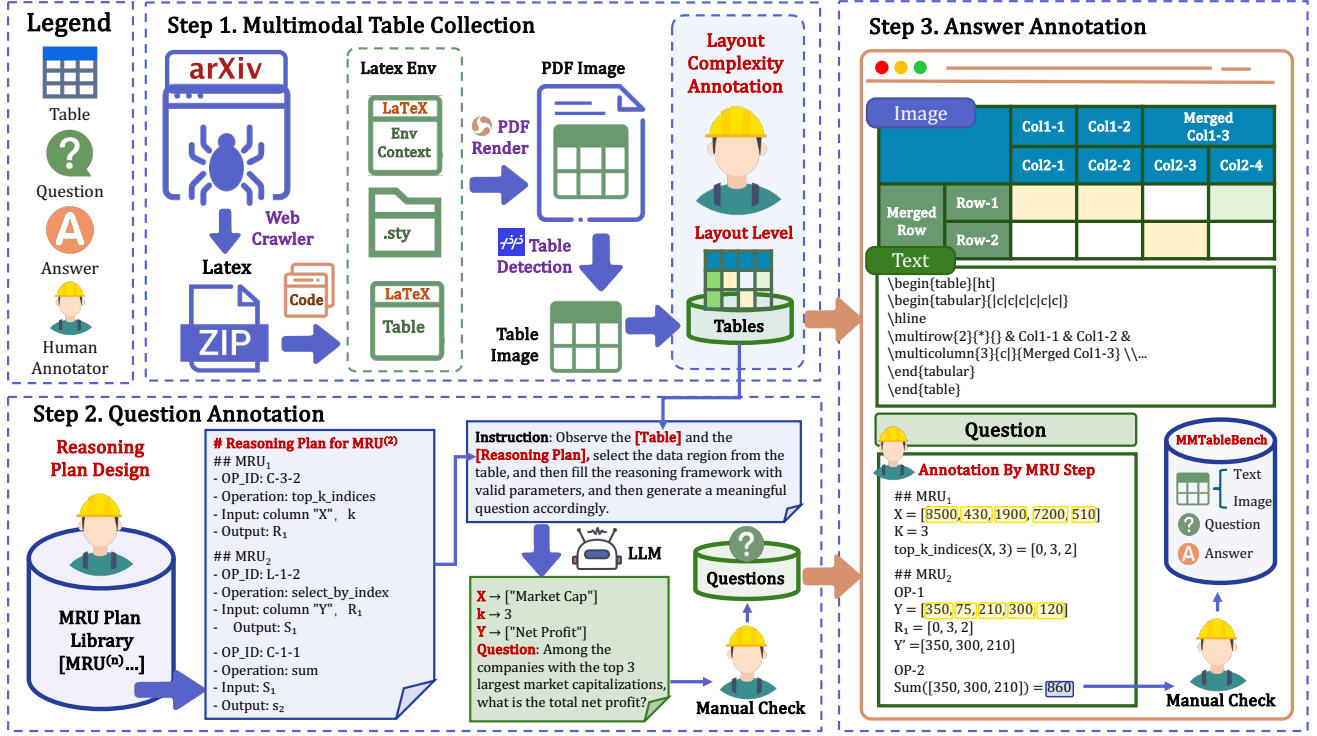


Figure 2: Overview of the Construction of the MMTableBench Benchmark. Tables in both LaTeX and image formats are directly collected from the same arXiv sources. The annotation process covers both layout and reasoning complexity, supported by human–LLM collaboration for question generation and MRU-based answer annotation.

modalities, better preserving the structural fidelity and semantic richness of the table.

### 3 Construction of MMTableBench

To advance TableQA from superficial task difficulty toward interpretable and modular modeling of specific capabilities, we construct MMTableBench with a dual emphasis on layout and reasoning complexity. To achieve this, we propose a novel annotation framework that (i) ensures faithful alignment between table images and their LaTeX counterparts, (ii) enables fine-grained and controllable annotations of both layout and reasoning complexity, and (iii) incorporates step-wise supervision with verifiable reasoning chains. These collectively form a unified framework with controllable complexity and three-way alignment across visual tables, textual tables, and reasoning chains, as shown in Figure 2.

Consequently, MMTableBench comprises meticulously curated samples encompassing diverse table structures, complex reasoning chains, and aligned image–text pairs. Each sample is annotated along two orthogonal axes—layout and reasoning complexity—each stratified into five levels.

While existing benchmarks incorporate complex numerical reasoning and non-trivial layouts, they typically lack fine-grained and orthogonal annotations that isolate structural from cognitive difficulty. This limitation hinders the diagnosis of whether model failures stem from layout understanding or reasoning ability. MMTableBench bridges this gap by enabling level-wise and

fine-grained assessment of layout understanding and reasoning competence, thus facilitating more interpretable and capability-specific evaluation of TableQA models, as summarized in Table 1.

#### 3.1 Multimodal Table Collection

To ensure realistic layout diversity and rich multimodal representations, we construct our dataset from scientific publications hosted on arXiv, which frequently contain complex and diverse table formats. The table collection pipeline is illustrated in Figure 2 and involves the following key steps:

**LaTeX-to-Image Table Construction.** We collect LaTeX source files from arXiv and extract all tabular environments along with the necessary compilation context. A minimal standalone LaTeX environment is then constructed by incorporating the extracted table code and associated external .sty files, allowing us to compile each table into an isolated PDF. The resulting PDF is rendered into an image, and we apply the PP-DoCLayOut [24] model to detect and crop the table region. This process yields a clean table-only image paired with its corresponding LaTeX source.

**Layout Categories.** We propose a scoring framework that systematically evaluates the complexity of table layouts along three orthogonal dimensions:

- **Header Structure (h):** Assesses if the table contains single-level or multi-level headers, either row-wise or column-wise.
- **Header Cell Spanning (s):** Measures the extent of cell merging across rows or columns.

**Table 1: Comparison of existing table QA benchmarks. MMTTableBench uniquely supports both complex and hierarchical layout structures, deep reasoning, and aligned image-text pairs.**

| Dataset                    | Numerical Reasoning | Hierarchical Layout | Multi-Level Reasoning | Complexity Layout | Image-Text Alignment | Table Format          |
|----------------------------|---------------------|---------------------|-----------------------|-------------------|----------------------|-----------------------|
| WTQ [21]                   | ✗                   | ✗                   | ✗                     | ✗                 | ✗                    | CSV                   |
| TabFact [3]                | ✗                   | ✗                   | ✗                     | ✗                 | ✗                    | CSV                   |
| FinQA [5]                  | ✓                   | ✗                   | ✗                     | ✗                 | ✗                    | JSON                  |
| TAT-QA [39]                | ✓                   | ✗                   | ✗                     | ✗                 | ✗                    | JSON                  |
| AIT-QA [13]                | ✓                   | ✓                   | ✗                     | ✗                 | ✗                    | JSON                  |
| HiTab [7]                  | ✗                   | ✓                   | ✗                     | ✗                 | ✗                    | JSON                  |
| BIRD [17]                  | ✓                   | ✗                   | ✗                     | ✗                 | ✗                    | Database              |
| SpreadSheetBench [19]      | ✗                   | ✓                   | ✗                     | ✗                 | ✗                    | EXCEL                 |
| TableBench [31]            | ✓                   | ✗                   | ✗                     | ✗                 | ✗                    | JSON                  |
| TableVQA-Bench [14]        | ✗                   | ✗                   | ✗                     | ✗                 | ✓                    | Image + Markdown/HTML |
| ComTQA [37]                | ✗                   | ✗                   | ✗                     | ✗                 | ✗                    | Image                 |
| MMTab [38]                 | ✓                   | ✓                   | ✗                     | ✗                 | ✓                    | Image + JSON          |
| RealHiTBench [29]          | ✓                   | ✓                   | ✗                     | ✗                 | ✓                    | Image + LaTeX/HTML    |
| MTabVQA [22]               | ✓                   | ✓                   | ✗                     | ✗                 | ✗                    | Image                 |
| <b>MMTableBench (Ours)</b> | ✓                   | ✓                   | ✓                     | ✓                 | ✓                    | Image + LaTeX         |

**Table 2: Structure identifiers, layout types, and assigned complexity scores used in the layout complexity scoring system.**

| Structure Dimension  | Structure ID | Score | Layout Type                                | Description                                            |
|----------------------|--------------|-------|--------------------------------------------|--------------------------------------------------------|
| Header Structure     | hs_001       | 0     | single-level row-header layout             | Only the first column serves as the row header         |
|                      | hs_002       | 0     | single-level column-header layout          | Only the first row serves as the column header         |
|                      | hs_003       | 1     | multi-level row-header layout              | Hierarchical row headers only                          |
|                      | hs_004       | 1     | multi-level column-header layout           | Hierarchical column headers only                       |
|                      | hs_005       | 1     | single-row and single-column header layout | Both row and column headers exist, each with one level |
|                      | hs_006       | 2     | single-row and multi-column header layout  | One row header with multi-level column headers         |
|                      | hs_007       | 2     | multi-row and single-column header layout  | One column header with multi-level row headers         |
|                      | hs_008       | 3     | multi-row and multi-column header layout   | Hierarchical headers along both row and column axes    |
| Header Cell Spanning | hcs_001      | 0     | no-spanning layout                         | No merged header cells                                 |
|                      | hcs_002      | 1     | row-header vertical spanning layout        | Row headers span multiple rows                         |
|                      | hcs_003      | 1     | column-header horizontal spanning layout   | Column headers span multiple columns                   |
|                      | hcs_004      | 2     | mixed spanning layout                      | Both row and column headers include merged cells       |
| Composite Structure  | ct_001       | 0     | standalone layout                          | A single complete and self-contained layout            |
|                      | ct_002       | 1     | flat composite layout                      | Multiple sub-layouts arranged side by side             |
|                      | ct_003       | 2     | nested composite layout                    | Layouts embedded within cells of another layout        |

- **Composite Structure (c):** Describes the overall structural composition, such as standalone, flat composite, or nested composite tables.

**Layout Complexity.** Each table is annotated with predefined structural dimensions, each consisting of multiple structure types associated with fixed complexity scores, as detailed in Table 2. The overall layout complexity is computed as the sum of three dimension-wise scores, represented by the triplet  $(h, s, c)$ :

$$\mathcal{L}_{lvi} = l(\mathcal{L}_{score}), \quad \mathcal{L}_{score} = h + s + c, \quad (3)$$

where the mapping function  $l(\cdot)$  converts the total score into one of five layout complexity levels, based on the intervals defined in Table 4 in Appendix B.1. This provides a quantitative and interpretable assessment of structural difficulty.

**Balanced Layout Complexity Selection.** We select tables to capture both structural diversity and a balanced distribution of complexity, covering layouts ranging from simple flat forms to deeply nested and composite structures. MMTTableBench systematically covers the full spectrum of structural complexity, offering a more comprehensive and balanced benchmark than existing datasets.

### 3.2 Question Annotation

To systematically control the reasoning complexity of questions in MMTTableBench, we design a structured question annotation process (as shown in Step 2 of Figure 2) that explicitly represents multi-step reasoning.

**Micro Reasoning Unit (MRU).** Fine-grained control over reasoning complexity in MMTTableBench is realized through the construction of questions from explicit symbolic reasoning plans, as opposed to open-end natural language generation. Each plan consists of a

structured sequence of MRUs, where each MRU denotes an indivisible logical operation that forms a complete atomic reasoning step. Specifically, the  $i$ -th MRU is defined as:

$$\text{MRU}_i = [L_i^{(1)}, L_i^{(2)}, \dots, L_i^{(k)}, C_i], \quad (4)$$

where  $L_i^{(k)}$  denotes the  $k$ -th table lookup operation, and  $C_i$  denotes a computation that is applied after all lookups have been executed. Importantly,  $C_i$  operates solely on the lookup results from  $L_i^{(1)}$  through  $L_i^{(k)}$ , thereby ensuring a clear execution order where reasoning proceeds from data retrieval to computation.

All operators are drawn from a predefined library of atomic operations, which is detailed in Table 5 in Appendix B.2 and comprises two categories:

- **Lookup operators**, which identify and return specific regions, cells, or entries from the table as lookup results;
- **Computation operators**, which perform computations on the outputs of lookup operators.

**Reasoning Plan.** A complete reasoning plan  $\text{MRU}^{(n)}$  is defined as a sequence of MRUs:

$$\text{MRU}^{(n)} = \text{MRU}_n \circ \text{MRU}_{n-1} \circ \dots \circ \text{MRU}_1, \quad (5)$$

where  $n$  denotes the number of valid reasoning units, and  $\circ$  represents the composition operator indicating that each  $\text{MRU}_i$  takes the output of the preceding one as its input. The special case of  $\text{MRU}^{(0)}$  corresponds to the lowest level of complexity, involving only atomic lookup operations without any computation.

**Reasoning Complexity.** The reasoning complexity is determined by the number of valid MRUs in the plan:

$$\mathcal{R}_{\text{Lvi}} = r(\mathcal{R}_{\text{score}}), \quad \mathcal{R}_{\text{score}}(\text{MRU}^{(n)}) = n, \quad (6)$$

where  $n$  denotes the total number of MRUs in the reasoning plan, and the mapping function  $r(\cdot)$  converts the raw reasoning score into a discrete complexity level, as defined in Table 6 in Appendix B.2.

**Reasoning Plan Library Construction.** Before question generation, we design a library of multi-step reasoning plans by manually composing atomic MRU operators into semantically coherent and computationally valid combinations. This library encodes diverse symbolic reasoning patterns independent of specific table content, enabling abstract logical structures to be reused across different instances. Each plan defines a symbolic reasoning path that can later be instantiated with concrete data.

**Question Construction.** Given the predefined reasoning plan library, each plan is instantiated with specific table content—filling in column names, row entities, and numerical values—to form executable MRU instances. Based on these instantiated reasoning paths, we employ LLMs to generate corresponding natural language questions. The model is guided to produce questions that implicitly convey the intended reasoning while avoiding explicit disclosure of intermediate computational steps. Finally, all generated questions are manually verified to ensure fluency, fidelity to the underlying reasoning logic, and clarity for human interpretation.

### 3.3 Answer Annotation

Answer annotation is achieved through a plan-driven annotation workflow (as shown in Step 3 of Figure 2), where each question is paired with a structured MRU plan that decomposes the task

into a sequence of MRUs. For each MRU, annotators select the relevant input region from the table image according to the specified parameters and choose an appropriate computation operator. The platform then automatically executes the operation to produce the intermediate result. This process iterates over all MRUs in the plan until the final answer is obtained.

In contrast to traditional end-to-end answer annotation, our approach provides fine-grained supervision over the entire reasoning process. Each annotated step explicitly specifies the input values, operation type, and output result in a standardized format, ensuring that the reasoning plan remains interpretable, executable, and verifiable. By offloading computation to the system, this framework simplifies the annotation workflow while preserving all intermediate reasoning steps, enabling full traceability and step-wise correction.

### 3.4 Quality Control

We engage eight undergraduate students and three expert annotators, all proficient in Excel for table manipulation, to participate in our study. Each annotator receives a comprehensive annotation guideline and operates within a streamlined web-based annotation platform. Regular coordination meetings are conducted to ensure consistent alignment throughout the annotation process.

**Layout Annotation Verification.** To ensure high-quality layout annotation, each structural dimension is independently labeled by at least two annotators, with conflicts resolved through adjudication. Cross-table consistency checks are additionally conducted to ensure that structurally similar layouts are assigned the same layout type. **Question and Answer Alignment.** Each natural language question undergoes manual review to ensure semantic alignment with its corresponding reasoning plan. Annotators confirm that the question faithfully reflects the intended logic while avoiding explicit disclosure of intermediate computational steps.

Final answers are verified through a two-stage process involving human annotation and system-level validation. Annotators are responsible only for performing necessary value lookups from the table, while all arithmetic and logical operations are executed by the annotation system. This system further verifies operator-specific input-output constraints to ensure the validity of each reasoning step. This rigorous pipeline ensures annotation consistency, correctness, and scalability throughout the benchmark.

## 4 Experiments

### 4.1 Implementation Details

We adopt the OpenAI GPT-o4-mini model as the supporting LLM for the annotation process. For experimental evaluation, we run open-source models using the Huggingface Transformers library on an A100 GPU with 80GB memory, providing sufficient computational resources for large context windows and high-resolution image inputs. For closed-source models, we access their official APIs (e.g., OpenAI, Anthropic, Google Cloud) and follow the recommended configurations to ensure fair and reproducible comparisons. For all models, we set the temperature parameter to 0, when supported, to enforce deterministic decoding.

We adopt a unified prompt structure for both image-based and text-based inputs, differing solely in the format used to present the table (image versus text), to ensure consistency in multimodal

**Table 3: Performance of various models across reasoning complexity levels ( $\mathcal{R}_{lv1}$ – $\mathcal{R}_{lv5}$ ) and layout complexity levels ( $\mathcal{L}_{lv1}$ – $\mathcal{L}_{lv5}$ ) using Exact Match (EM). Bold values indicate the best performance for text input within each model category. Underlined values indicate the best performance for image input within each model category.**

| Model                         | Input | Reasoning Depth     |                     |                     |                     |                     | Layout Complexity   |                     |                     |                     |                     | Overall      |
|-------------------------------|-------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------|
|                               |       | $\mathcal{R}_{lv1}$ | $\mathcal{R}_{lv2}$ | $\mathcal{R}_{lv3}$ | $\mathcal{R}_{lv4}$ | $\mathcal{R}_{lv5}$ | $\mathcal{L}_{lv1}$ | $\mathcal{L}_{lv2}$ | $\mathcal{L}_{lv3}$ | $\mathcal{L}_{lv4}$ | $\mathcal{L}_{lv5}$ |              |
| Table-oriented Model          |       |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |              |
| Table-LLaVA-v1.5-7B           | Image | 1.01                | 4.03                | 4.03                | 3.58                | 2.29                | 2.01                | 4.08                | 2.80                | 4.37                | 1.75                | 3.00         |
|                               | Text  | 8.39                | 5.37                | 5.37                | 4.03                | 3.05                | 4.70                | 2.72                | 6.99                | 5.52                | 6.64                | 5.30         |
| Open-source Models            |       |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |              |
| Gemma-3-4B-Instruct           | Image | 8.72                | 11.41               | 3.36                | 2.68                | 0.00                | 4.03                | 2.04                | 6.99                | 7.59                | 6.29                | 5.36         |
|                               | Text  | 8.72                | 11.41               | 11.41               | 6.71                | 4.58                | 6.71                | 4.08                | 14.69               | 13.10               | 4.90                | 8.67         |
| InternVL3-8B                  | Image | 38.26               | 18.79               | 3.36                | 4.70                | 2.29                | 12.08               | 9.52                | 14.69               | 15.86               | 16.78               | 13.76        |
|                               | Text  | 24.50               | 9.40                | 3.36                | 4.25                | 0.76                | 10.07               | 7.82                | 9.79                | 8.97                | 6.53                | 8.64         |
| Qwen2-VL-7B-Instruct          | Image | 39.60               | 19.46               | 6.71                | 7.38                | 2.29                | 12.75               | 13.61               | 18.88               | 16.55               | 15.38               | 15.41        |
|                               | Text  | 34.23               | 22.15               | 10.07               | 8.72                | 1.53                | 16.78               | 12.93               | 18.18               | 17.24               | 13.29               | 15.68        |
| Qwen2.5-VL-7B-Instruct        | Image | 65.77               | 38.26               | 26.85               | 18.12               | 8.40                | 32.21               | 31.29               | 36.36               | 34.48               | 25.87               | 32.05        |
|                               | Text  | 61.07               | 28.19               | 22.15               | 17.45               | 5.34                | 25.50               | 27.21               | 27.27               | 33.10               | 23.78               | 27.37        |
| InternVL3-14B                 | Image | 69.80               | 46.15               | 32.21               | 21.70               | 10.69               | 42.28               | 35.71               | 39.16               | 37.47               | 29.37               | 36.83        |
|                               | Text  | 70.47               | 45.64               | 26.17               | 20.36               | 6.11                | 38.26               | 34.69               | 33.57               | 35.17               | 30.30               | 34.43        |
| Llama-3.2-90B-Vision-Instruct | Image | 42.95               | 33.56               | 14.09               | 10.07               | 7.63                | 23.49               | 21.09               | 21.68               | 20.69               | 23.08               | 22.01        |
|                               | Text  | 61.74               | 35.57               | 19.46               | 23.49               | 14.50               | 38.26               | 30.61               | 30.77               | 34.48               | 22.38               | 31.36        |
| Qwen2.5-VL-32B-Instruct       | Image | 71.14               | <u>60.40</u>        | 41.61               | 29.75               | <u>20.61</u>        | 47.65               | 46.94               | <u>50.35</u>        | <u>44.37</u>        | 37.06               | 45.30        |
|                               | Text  | 76.51               | 58.39               | 41.61               | <b>39.15</b>        | 15.27               | 52.35               | 48.98               | <b>49.65</b>        | 45.06               | <b>38.46</b>        | 46.95        |
| Qwen2.5-VL-72B-Instruct       | Image | <u>75.17</u>        | 55.70               | <u>46.31</u>        | <u>34.23</u>        | 19.08               | <u>50.34</u>        | <u>48.98</u>        | 48.95               | 44.14               | <u>41.26</u>        | <u>46.77</u> |
|                               | Text  | <b>78.52</b>        | <b>59.73</b>        | <b>44.30</b>        | 33.11               | <b>21.37</b>        | <b>53.02</b>        | <b>51.02</b>        | 48.95               | <b>52.64</b>        | 34.27               | <b>48.05</b> |
| Closed-source Models          |       |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |              |
| Qwen-VL-Max                   | Image | 70.47               | 60.40               | 43.62               | 30.87               | 23.66               | 49.66               | 47.62               | 51.05               | 44.14               | 39.16               | 46.35        |
|                               | Text  | 77.18               | 59.73               | 46.98               | 40.27               | 23.66               | 55.70               | 52.38               | 52.45               | 51.72               | 38.46               | 50.21        |
| Gemini-2.5-Flash              | Image | 77.85               | 57.05               | 41.61               | 34.23               | 28.24               | 53.02               | 45.58               | 46.85               | 48.97               | 46.85               | 48.28        |
|                               | Text  | 80.54               | 61.07               | 38.93               | 36.24               | 27.48               | 51.68               | 53.74               | 52.45               | 46.21               | 42.66               | 49.38        |
| GPT-4.1-mini                  | Image | 75.50               | 59.73               | 46.31               | 38.48               | 27.48               | 57.05               | 50.34               | 52.45               | 47.13               | 43.01               | 50.05        |
|                               | Text  | 80.54               | 60.40               | 51.01               | 35.57               | 29.77               | 58.39               | 54.42               | 51.05               | 48.97               | 46.85               | 51.99        |
| Claude-Sonnet-4               | Image | 77.85               | <u>64.43</u>        | 49.66               | 43.85               | 30.53               | 57.72               | 58.39               | 54.55               | 51.95               | 46.15               | 53.83        |
|                               | Text  | 80.54               | 66.44               | 51.01               | 48.99               | 34.35               | 63.09               | 61.22               | 55.94               | 53.79               | 49.65               | 56.81        |
| GPT-o4-mini                   | Image | <u>79.19</u>        | 63.76               | 52.15               | 45.19               | 33.59               | 59.73               | <u>58.50</u>        | 55.94               | 50.57               | <u>51.75</u>        | 55.34        |
|                               | Text  | 80.54               | 67.79               | <b>55.03</b>        | 48.32               | 35.11               | 61.07               | 60.54               | 58.74               | <b>55.86</b>        | <b>53.15</b>        | 57.91        |
| Doubao-Seed-1.6               | Image | 78.52               | 63.76               | <u>52.35</u>        | <u>45.86</u>        | <u>36.64</u>        | <u>63.09</u>        | 56.46               | <u>58.74</u>        | <u>52.64</u>        | 48.25               | <u>55.89</u> |
|                               | Text  | <b>81.21</b>        | <b>69.80</b>        | 52.35               | <b>51.23</b>        | <b>38.17</b>        | <b>64.43</b>        | <b>63.95</b>        | <b>59.44</b>        | 54.02               | <b>53.15</b>        | <b>59.06</b> |

evaluation. All prompts follow the Chain-of-Thought (CoT) [27] paradigm to guide models through multi-step reasoning. To reduce the influence of irrelevant or verbose outputs, we impose formatting constraints on model responses and extract final answers using a standardized parsing procedure.

## 4.2 LLMs

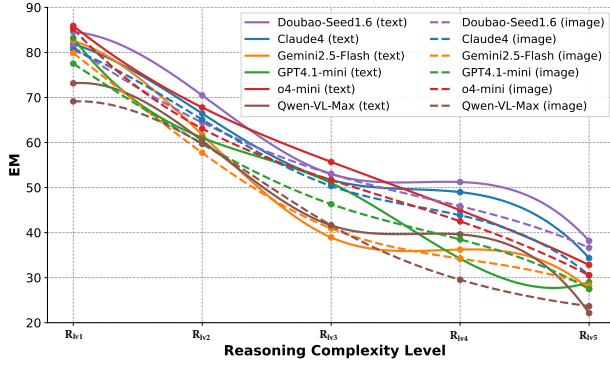
We evaluate 15 advanced multimodal large language models (MLLMs) on MMTableBench, spanning parameter sizes from 4B to 90B and covering both open-source and proprietary variants. The evaluation covers the table-oriented model Table-LLaVA [38]; open-source models including Gemma [26], InternVL [6], Llama3-Vision [10],

Qwen2-VL, and Qwen2.5-VL [33]; as well as closed-source models such as GPT-4.1-mini, GPT-o4-mini [1], Gemini-2.5-flash, Claude-Sonnet-4, Qwen-VL-Max [33], and Doubao-Seed-1.6.

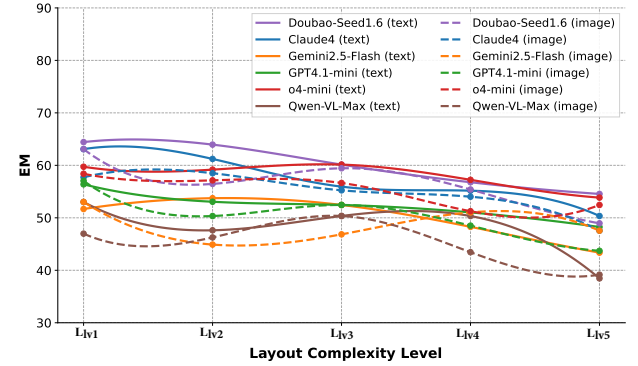
## 4.3 Metrics

We adopt Exact Match (EM) as the primary evaluation metric for numerical computation tasks. Given a table-question pair with ground-truth answers  $A = \{a_1, \dots, a_k\}$  and model predictions  $\hat{A} = \{\hat{a}_1, \dots, \hat{a}_k\}$ , the EM score is defined as

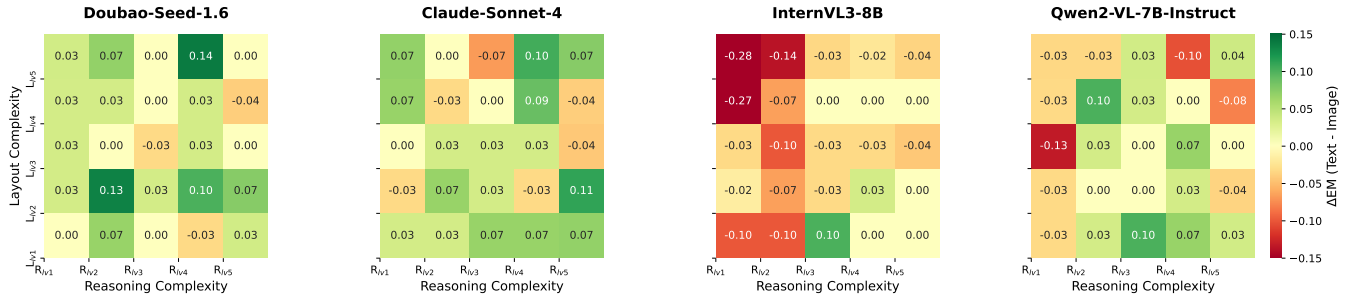
$$\text{EM}(A, \hat{A}) = \frac{1}{k} \sum_{i=1}^k \mathbb{I}(a_i = \hat{a}_i), \quad (7)$$



(a) Effect of reasoning complexity on model performance.



(b) Effect of layout complexity on model performance.

**Figure 3: Model performance trends under different types of complexity: (a) reasoning complexity; (b) layout complexity.****Figure 4: Heatmaps illustrate the accuracy differences ( $\Delta EM = \text{text}_{EM} - \text{image}_{EM}$ ) for four multimodal models over combinations of layout complexity ( $\mathcal{L}_{lv1} - \mathcal{L}_{lv5}$ ) and reasoning complexity ( $\mathcal{R}_{lv1} - \mathcal{R}_{lv5}$ ). The color scale reflects the magnitude and direction of the change, with green indicating accuracy gains and red indicating drops.**

where  $\mathbb{I}(\cdot)$  is the indicator function that equals 1 when the prediction exactly matches the ground truth and 0 otherwise. To account for numerical precision issues, we additionally report relaxed EM variants with tolerance thresholds (e.g., 2%, 5%, and 10%) in Appendix C.1. Unless otherwise specified, all results in the main paper are based on strict EM with zero tolerance.

#### 4.4 Main Results

We systematically evaluate advanced multimodal large language models (MLLMs) on MMTableBench across two critical dimensions: reasoning complexity ( $\mathcal{R}_{lv1} - \mathcal{R}_{lv5}$ ) and layout complexity ( $\mathcal{L}_{lv1} - \mathcal{L}_{lv5}$ ), as shown in Table 3. For each model, both image and text inputs are assessed under identical task settings, enabling direct comparison of modality-specific performance.

**Overall Performance Gap.** Closed-source models consistently outperform their open-source counterparts in both reasoning and layout dimensions. Among all evaluated models, Doubao-Seed-1.6 achieves the highest overall accuracy, followed by GPT-o4-mini and Claude-Sonnet-4. In contrast, open-source models generally underperform, with Qwen2.5-VL-72B-Instruct achieving the best performance within this group. Table-oriented models exhibit clear performance limitations, suggesting a bottleneck in their ability to handle complex reasoning and layout challenges.

**Impact of Reasoning Complexity.** Model performance deteriorates significantly as reasoning depth increases, underscoring the inherent challenge posed by complex reasoning tasks. For instance, the performance of Doubao-Seed-1.6 drops sharply from  $\mathcal{R}_{lv1}$  to  $\mathcal{R}_{lv5}$ . This consistent trend is observed across all models, confirming that reasoning complexity constitutes a significant performance bottleneck, even for top-performing MLLMs.

**Impact of Layout Complexity.** As layout complexity increases, all models exhibit a declining performance trend, indicating that intricate layouts impede the understanding of table structure. Nevertheless, the performance of Doubao-Seed-1.6 decreases only slightly from  $\mathcal{L}_{lv1}$  to  $\mathcal{L}_{lv5}$ , which is substantially smaller than the performance loss caused by reasoning complexity. Despite the structural complexity and visual ambiguity introduced by intricate layouts, the models maintain reliable layout awareness and spatial reasoning capabilities. Interestingly, some models (especially open-source ones) perform better on moderately complex layouts (e.g.,  $\mathcal{L}_{lv3}$  or  $\mathcal{L}_{lv4}$ ), showing a non-monotonic trend. This may be due to the richer structural cues in such layouts and their higher prevalence in training data, which together enhance model performance.

**Modal Discrepancy.** Most multimodal large language models perform better with textual input, indicating the continued dominance of the text modality, whereas some open-source models exhibit

stronger performance with image input. This difference reflects divergent multimodal optimization strategies between proprietary and open-source models.

## 5 Further Analysis

### RQ1: How do reasoning and layout complexity respectively challenge multimodal models in TableQA?

To comprehensively investigate the impact of different dimensions of complexity on model performance, we conduct a comparative analysis under varying levels of reasoning depth and structural layout complexity, as illustrated in Figure 3. The results show that both complexity factors lead to performance degradation, albeit to varying degrees. As reasoning depth increases, model accuracy declines sharply, highlighting the inherent challenge posed by deep reasoning tasks. In contrast, increased layout complexity results in only moderate performance drops, suggesting that models are comparatively more robust to structural variations.

A further modality-specific analysis reveals consistent sensitivity to reasoning depth across both image and text inputs, as illustrated in Figure 3(a). With respect to layout complexity (Figure 3(b)), text-based inputs—represented by the solid line—achieve generally higher accuracy but exhibit a pronounced performance decline as structural complexity increases, indicating limited robustness to layout variation. In contrast, image-based inputs—depicted by the dashed line—maintain a more stable, undulating performance trend despite consistently lower absolute accuracy, reflecting stronger resilience to complex structural formats. These findings suggest that while visual inputs offer limited benefits for semantic reasoning, they provide complementary advantages in handling structurally complex tables by enhancing layout perception and adaptability.

### RQ2: Do multimodal models exhibit the expected modality specialization as task complexity increases?

A central motivation behind the design of MLLMs is to mitigate the structural information loss inherent in text linearization by incorporating visual input. Tables, as inherently two-dimensional structures, convey semantics through spatial layouts that often exceed the expressive capacity of linear text. These models are thus expected to exhibit modality-specific behavior during task execution—leveraging the visual modality to better perceive layout structures while relying on the language modality to perform logical reasoning. Based on the MMTTableBench evaluation framework, we investigate modality behavior along two key dimensions: layout complexity and reasoning complexity. Figure 4 presents the performance disparities between modalities across various combinations of structural and reasoning difficulty.

We find that advanced multimodal models—such as Doubao-Seed-1.6 and Claude-Sonnet-4—exhibit more balanced performance across visual and textual modalities under varying combinations of reasoning and structural complexity. In contrast, open-source models, particularly those optimized for visual understanding (e.g., InternVL3-8B and Qwen2-VL-7B-Instruct), display more evident modality-specific divergence. As illustrated in the upper-left region of the heatmap—corresponding to tasks with low reasoning complexity but high structural complexity—a higher concentration of red cells indicates that image-based inputs yield superior performance, likely due to enhanced structural perception. Conversely,

the lower-right region—characterized by high reasoning demands and simpler layouts—shows a prevalence of green areas, suggesting that textual inputs remain more effective for deep logical reasoning, where visually grounded models continue to underperform.

## 6 Related Work

Table Question Answering (TableQA) plays a vital role in enabling intelligent systems to understand and reason over structured information [32, 35, 40]. With the rapid advancement of large language models (LLMs), this field has witnessed remarkable progress, expanding its scope from simple textual tables to complex, multimodal scenarios [2, 23, 28, 30, 34, 36]. Early benchmarks, including WikiTableQuestions [21], TabFact [3], SQA [11], HybridQA [4], and FeTaQA [20], primarily focus on textual tables and support tasks such as fact verification, multi-hop reasoning, and free-form answer generation. Subsequent datasets, such as TAT-QA [39], FinQA [5], BizBench [15], and AIT-QA [13], further introduce numerical reasoning grounded in financial documents. However, most of these benchmarks rely on clean, flattened HTML tables, where linearized representations obscure original structural semantics and limit applicability to real-world scenarios. To better reflect realistic table understanding, recent work has extended TableQA to the multimodal domain by incorporating table images and layout-aware reasoning. Benchmarks such as MultiModalQA [25], MMTTab [38], RealHiT-Bench [29], and TableVQA-Bench [14] introduce challenges including irregular layouts, OCR noise, and visual-text misalignment. In parallel, both proprietary and open-source multimodal models have demonstrated strong performance in multimodal table reasoning, showing notable progress in visual-structural understanding. Nevertheless, despite these advances, existing benchmarks intertwine reasoning complexity with layout difficulty. This coupling obscures the diagnosis of model failures and *makes it difficult to determine whether errors stem from logical reasoning gaps or structural perception deficiencies*. By independently varying reasoning and layout complexity dimension, MMTTableBench facilitates fine-grained evaluation and offers deeper insights into the limitations of MLLMs in realistic table reasoning scenarios.

## 7 Conclusion

To enable fine-grained, capability-oriented evaluation of multimodal TableQA, we introduce MMTTableBench, a multi-level benchmark that disentangles layout complexity from reasoning depth. We further propose a unified annotation framework aligning visual tables, textual structures, and step-wise reasoning with controllable complexity. Based on MMTTableBench, we conduct comprehensive experiments on advanced MLLMs to analyze their performance limits under diverse structural and reasoning demands, providing a principled basis for understanding cross-modal table reasoning.

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## A Motivation

### For what purpose was the dataset created?

MMTableBench is designed to assess the capabilities of Multimodal Large Language Models (MLLMs) on table question answering (Table QA) tasks involving visual table inputs. Its design is motivated by two key limitations observed in existing benchmarks:

- **Prevalence of visual tables in practical scenarios:** In real-world settings, tables are often embedded in formats such as PDFs, scanned documents, or screenshots, which are not machine-readable and thus present significant challenges for Table QA systems.
- **Entanglement of reasoning and layout complexity in existing benchmarks:** Many current datasets conflate multiple factors—such as reasoning depth and structural layout—making it difficult to conduct targeted evaluations and pinpoint specific model weaknesses.

*We argue that a well-designed benchmark should not merely present challenges, but also expose model limitations and provide diagnostic insights.* MMTableBench disentangles reasoning and layout complexity into two orthogonal dimensions, enabling fine-grained evaluation of how models process hierarchical layouts and complex reasoning chains. This design further reveals the trade-offs and biases between visual perception and reasoning capabilities in multimodal large language models (MLLMs).

### Why Are Reasoning and Structural Complexity Chosen as the Core Evaluation Dimensions?

We identify reasoning complexity and layout complexity as two orthogonal and critical factors shaping multimodal TableQA performance. Both are central to advancing multimodal large language models (MLLMs).

Layout complexity stems from the spatial structure of tables—such as multi-level headers and merged cells—which goes beyond linear token sequences and challenges both symbolic and neural approaches. Even humans rely on visual cues for understanding, highlighting the need to model structure explicitly. Reasoning complexity reflects the challenge of performing multi-step logical inference over semi-structured data. Despite LLMs’ progress in language tasks, they struggle with table-specific operations like aggregation, comparison, and multi-hop reasoning—capabilities essential for real-world analytical tasks.

By disentangling these two dimensions, MMTableBench enables fine-grained, interpretable evaluation of model strengths and limitations across structural and reasoning axes.

## B Details of Layout and Reasoning Complexity

### B.1 Layout Complexity Details

Table 2 provides a comprehensive summary of all structural types considered in our framework, along with their corresponding descriptions and assigned scores. Each structural dimension—such as header hierarchy, cell merging, and composite organization—is quantified to capture its contribution to overall layout complexity. The cumulative score, ranging from 0 (least complex) to 8 (most complex), reflects the aggregated structural difficulty of a table. To

facilitate consistent interpretation and analysis, these continuous scores are further discretized into five levels, as detailed in Table 4.

### B.2 Reasoning Complexity Details

To enable systematic multi-step reasoning over visual tables, we define a unified library of atomic operators that serve as the foundational components of Micro Reasoning Units (MRUs). Each atomic operator performs a single, clearly specified transformation on tabular data, and MRUs are constructed by sequentially composing these operations. The operator set is categorized into six types: aggregation, scalar arithmetic, selection, row-wise arithmetic, row lookup, and index manipulation. Table 5 summarizes all atomic operators, detailing their categories, input/output signatures, and example behaviors.

As shown in Table 6, we categorize reasoning complexity into five levels according to the number of MRUs involved.  $MRU^{(0)}$  serves as a special case, consisting solely of lookup operations without any computation; therefore, it is not considered a valid MRU, and its reasoning complexity is defined as zero.

**Table 4: Five levels of layout complexity based on total structure scores.**

| Layout Level        | Score Range | Description                          |
|---------------------|-------------|--------------------------------------|
| $\mathcal{L}_{lv1}$ | 0           | Minimal structural complexity        |
| $\mathcal{L}_{lv2}$ | 1–2         | Simple and regular structure         |
| $\mathcal{L}_{lv3}$ | 3–4         | Moderate structural complexity       |
| $\mathcal{L}_{lv4}$ | 5–6         | Intricate or irregular structure     |
| $\mathcal{L}_{lv5}$ | 7           | Highly complex and layered structure |

## C Supplementary Experimental Results

### C.1 Experimental Results of EM Variants

Figure 5 shows how varying error-tolerant thresholds affect the overall performance of closed-source models, using exact match (EM) metrics at 0%,  $\pm 2\%$ ,  $\pm 5\%$ , and  $\pm 10\%$  tolerance levels.

Performance consistently improves across all models as the tolerance increases, suggesting that many strict EM mismatches are due to minor numerical deviations rather than substantive errors.  $EM_{\pm 2}$  and  $EM_{\pm 5}$  recover most of these near-miss cases, alleviating the harshness of strict exact match evaluation. Despite these absolute gains, the relative model rankings remain largely unchanged, indicating that EM-based metrics, regardless of tolerance, reliably reflect comparative model quality. Thus, threshold choice impacts score magnitude but not evaluation consistency.

### C.2 MLLM Performance Across Reasoning and Layout Complexity Dimensions

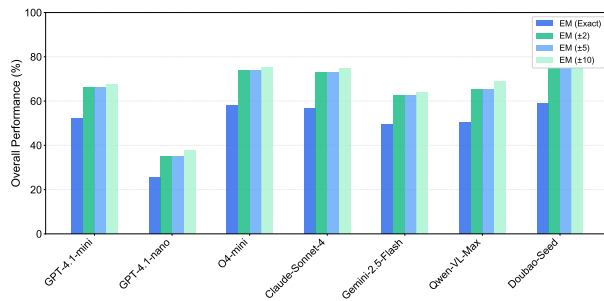
We provide two sets of heatmaps to analyze MLLM performance under different input modalities and task complexities. Figure 6 displays the absolute accuracy under text-only input, reflecting each model’s pure language reasoning capabilities. In contrast, Figure 7 shows the performance when models rely solely on image inputs, highlighting their visual understanding and layout parsing abilities.

**Table 5: MRU atomic operations library.**

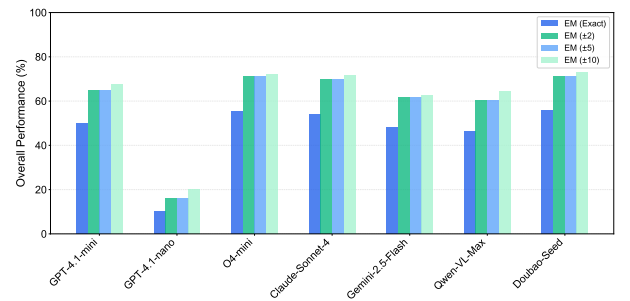
| ID    | Category            | Operator                 | Input Type                        | Output Type       | Semantics / Example                |
|-------|---------------------|--------------------------|-----------------------------------|-------------------|------------------------------------|
| C-1-1 | Aggregation         | sum                      | scalar sequence                   | scalar            | Summation,                         |
| C-1-2 | Aggregation         | mean                     | scalar sequence                   | scalar            | Arithmetic mean,                   |
| C-1-3 | Aggregation         | median                   | scalar sequence                   | scalar            | Median,                            |
| C-1-4 | Aggregation         | max                      | scalar sequence                   | scalar            | Maximum value                      |
| C-1-5 | Aggregation         | min                      | scalar sequence                   | scalar            | Minimum value                      |
| C-1-6 | Aggregation         | count                    | scalar sequence                   | integer scalar    | Element count                      |
| C-2-1 | Arithmetic          | add                      | scalar, scalar                    | scalar            | Addition                           |
| C-2-2 | Arithmetic          | diff                     | scalar, scalar                    | scalar            | Ordered subtraction                |
| C-2-3 | Arithmetic          | percent_diff             | scalar, scalar                    | percentage scalar | Relative percentage change         |
| C-2-4 | Arithmetic          | multiply                 | scalar, scalar                    | scalar            | Multiplication                     |
| C-2-5 | Arithmetic          | divide                   | scalar, scalar                    | scalar            | Ordered division                   |
| C-2-6 | Arithmetic          | ratio                    | scalar, scalar                    | scalar            | Ratio between two scalars          |
| C-3-1 | Selection           | top_k_vals               | scalar sequence, scalar           | scalar sequence   | Top- $k$ largest values            |
| C-3-2 | Selection           | top_k_indices            | scalar sequence, scalar           | index sequence    | Row indices of top- $k$ values     |
| C-3-3 | Selection           | least_k_vals             | scalar sequence, scalar           | scalar sequence   | Bottom- $k$ smallest values        |
| C-3-4 | Selection           | least_k_indices          | scalar sequence, scalar           | index sequence    | Row indices of bottom- $k$ values  |
| C-3-5 | Selection           | filter_col_vals          | scalar sequence, scalar, operator | scalar sequence   | Filter values satisfying condition |
| C-3-6 | Selection           | filter_row_indices       | scalar sequence, scalar, operator | index sequence    | Filtered row indices by condition  |
| C-4-1 | Row-wise Arithmetic | elementwise_multiply     | scalar sequence $\times 2$        | scalar sequence   | Row-wise product                   |
| C-4-2 | Row-wise Arithmetic | elementwise_divide       | scalar sequence $\times 2$        | scalar sequence   | Row-wise division                  |
| C-4-3 | Row-wise Arithmetic | elementwise_diff         | scalar sequence $\times 2$        | scalar sequence   | Row-wise difference                |
| C-4-4 | Row-wise Arithmetic | elementwise_ratio        | scalar sequence $\times 2$        | scalar sequence   | Row-wise ratio                     |
| C-4-5 | Row-wise Arithmetic | elementwise_percent_diff | scalar sequence $\times 2$        | scalar sequence   | Row-wise percent difference        |
| L-1-1 | Lookup              | get_col_val_by_index     | sequence, index                   | scalar            | Value at specific row index        |
| L-1-2 | Lookup              | select_by_index          | sequence, index sequence          | sequence          | Values at multiple row indices     |
| L-1-3 | Lookup              | row_intersect            | index sequence $\times 2$         | index sequence    | Set intersection of row indices    |

**Table 6: Examples of MRU reasoning plans.**

| MRU Plan           | Reasoning Level        | Score | Description                                                                                                                      | Example Reasoning Plan                                                                                                                                                                                                                                                                                               |
|--------------------|------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MRU <sup>(0)</sup> | $\mathcal{R}_{I_{o1}}$ | 0     | Lookup the value at a given row and column, based on structural or semantic filtering.                                           | MRU <sub>0</sub> : $s = \text{get\_col\_val\_by\_index}(\mathbb{S}_2, \mathbf{R})$                                                                                                                                                                                                                                   |
| MRU <sup>(1)</sup> | $\mathcal{R}_{I_{o2}}$ | 1     | Compute the sum over column $X$ , e.g., $X = [1, 2, 3]$ , result is 6.                                                           | MRU <sub>1</sub> : $s = \text{sum}(X)$                                                                                                                                                                                                                                                                               |
| MRU <sup>(2)</sup> | $\mathcal{R}_{I_{o3}}$ | 2     | Compute the mean of column $X$ , then subtract it from the $r$ -th row of column $Y$ .                                           | MRU <sub>1</sub> : $s_1 = \text{mean}(X)$<br>MRU <sub>2</sub> : $s_2 = s_{Y_r} - s_1$                                                                                                                                                                                                                                |
| MRU <sup>(3)</sup> | $\mathcal{R}_{I_{o4}}$ | 3     | Compute means of columns $\mathbb{S}_1$ and $\mathbb{S}_2$ , then calculate the difference.                                      | MRU <sub>1</sub> : $s_1 = \text{mean}(\mathbb{S}_1)$<br>MRU <sub>2</sub> : $s_2 = \text{mean}(\mathbb{S}_2)$<br>MRU <sub>3</sub> : $s = s_1 - s_2$                                                                                                                                                                   |
| MRU <sup>(4)</sup> | $\mathcal{R}_{I_{o5}}$ | 4     | Compute the mean of $\mathbb{S}_1$ as a threshold, filter $\mathbb{S}_2$ , average selected values, subtract from the threshold. | MRU <sub>1</sub> : $s_1 = \text{mean}(\mathbb{S}_1)$<br>MRU <sub>2</sub> : $\mathbf{R} = \text{filter\_row\_indices}(\mathbb{S}_2, s_1, \text{op})$ (e.g., $\text{op} = ">"$ )<br>MRU <sub>3</sub> : $s_2 = \text{mean}(\text{select\_by\_index}(\mathbb{S}_2, \mathbf{R}))$<br>MRU <sub>4</sub> : $s_3 = s_2 - s_1$ |

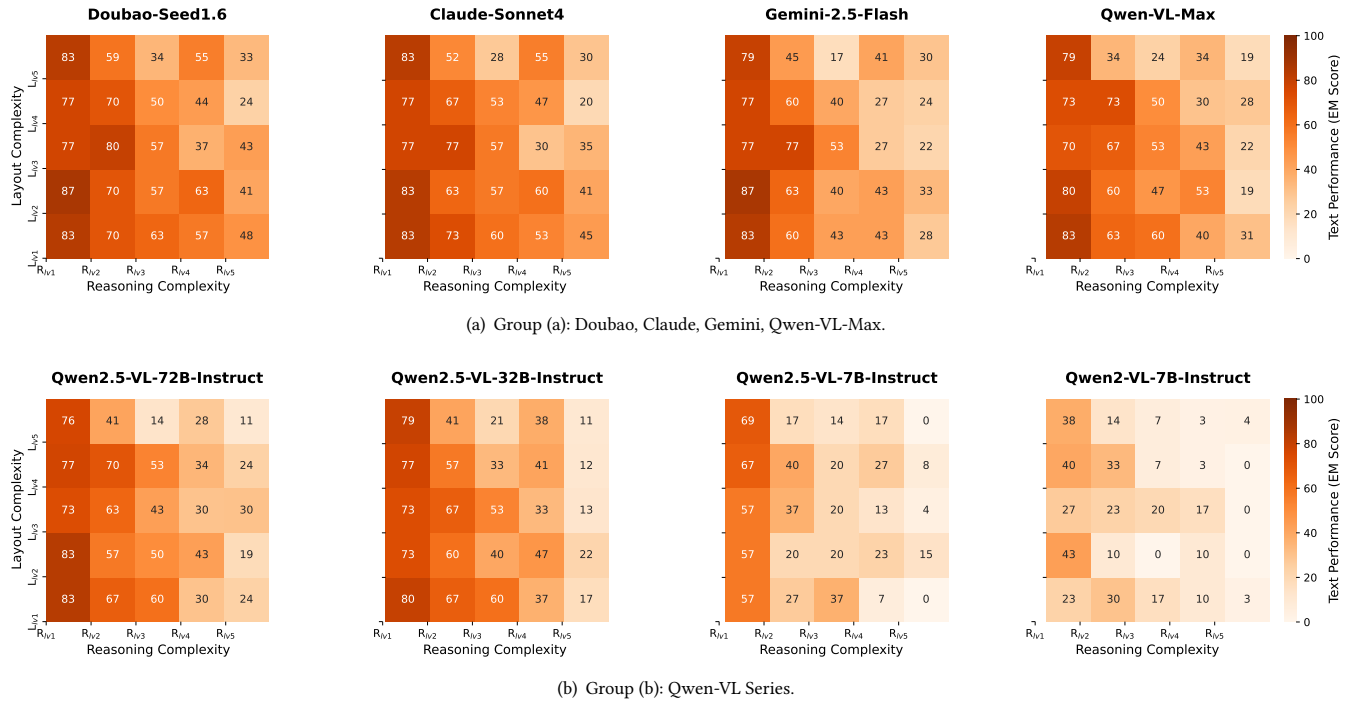


(a) Closed-Source Models Overall Performance (Text Input).

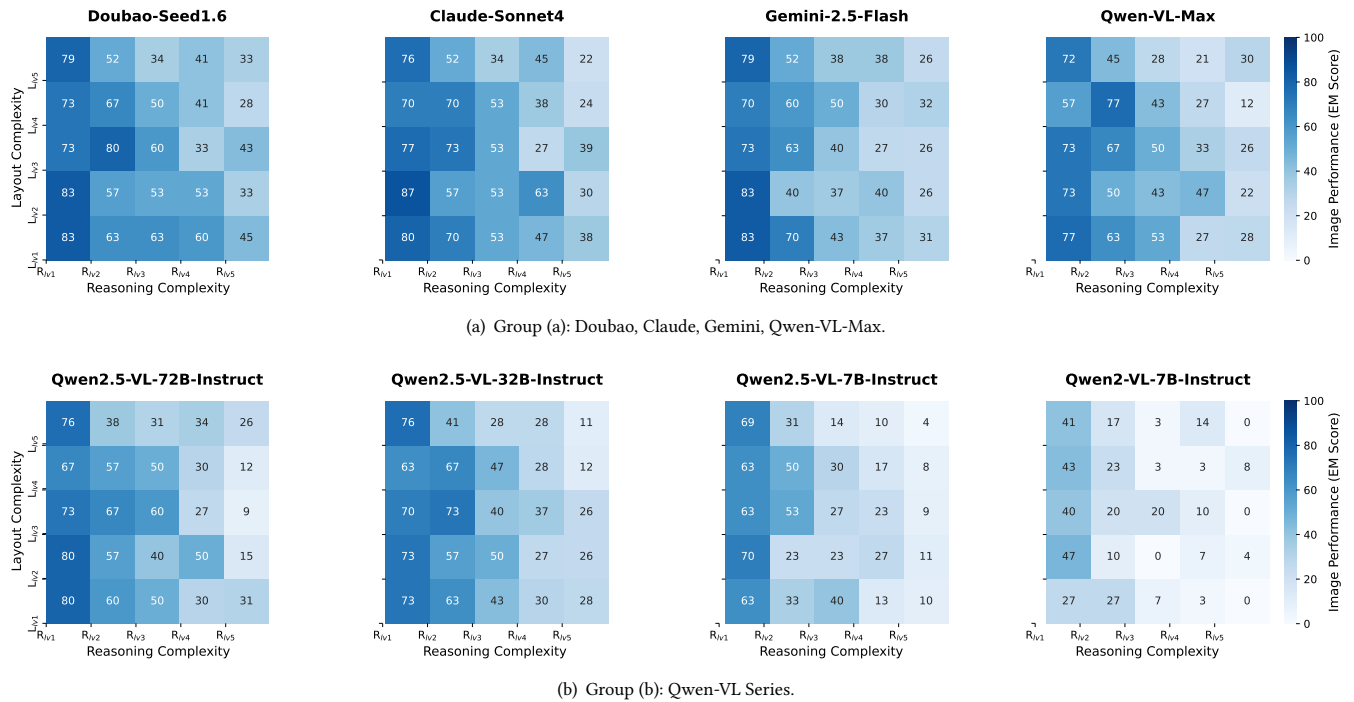


(b) Closed-Source Models Overall Performance (Image Input).

**Figure 5: Overall performance comparison of closed-source models across modalities: (a) text input; (b) image input.**



**Figure 6: Heatmaps of absolute performance (text input only) across layout and reasoning complexity levels for various models. Each cell represents the model's performance under a specific layout and reasoning complexity condition using text input.**



**Figure 7: Heatmaps of absolute performance (image input only) across layout and reasoning complexity levels for MLLMs. Each cell represents the model's performance under a specific layout and reasoning complexity condition using image input.**